

COSC 39: Theory of Computation

Problem 1. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

TRIANGLE

- **Input:** An undirected graph G
- **Output:** Does graph G contain a triangle?

Problem 2. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

MAXCLIQUE

- **Input:** An undirected graph G , a parameter k
- **Output:** Does graph G contain a clique of size at least k ?

Problem 3. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

PRIMEPOWER

- **Input:** An integer n , and a prime p
- **Output:** Is $n = p^k$ for some positive integer k ?

Problem 4. Put the following problem into the smallest complexity classes possible: P, NP, or merely computable?

COMPOSITES

- **Input:** An integer n
- **Output:** Is $n = pq$ for some positive integers $p, q > 1$?